Aerospace Technology

Annex, material after the first test

Notes by **Eden R.S.**

Master's Degree in Unmanned Aircraft Systems
and Associated Technologies · UPV

Academic year 2026-2027

Review answers for topics taught after unit 2.5 of the course (launch vehicles, surveillance systems, rotorcraft). This material is **not** in test 1 of 5 November 2026; it complements the main notes (`Apuntes_Tecnologia_Aeroespacial.pdf`).

Contents

8 Annex: material after the first test **2**

1 Annex: review answers on topics after unit 2.5 **3**

1.1 Mass estimate for single-stage and two-stage launch vehicles 3

1.2 Secondary radar, TCAS and ADS-B 4

1.3 The rotor in a few lines (introduction for Questions 18 to 20) 6

1.4 Vertical flight of a helicopter: climb and descent 6

1.5 Flight envelope of a helicopter 8

1.6 Yaw control of a tandem twin-rotor helicopter 9

UNIT 8

Annex: material after the first test

1 Annex: review answers on topics after unit 2.5

This annex holds the answers to review questions on topics taught **after unit 2.5** of the course (launch vehicles, surveillance systems, rotorcraft). They are **not** part of test 1 (5 November 2026); they are kept in this separate PDF so the main notes contain only test-1 material. Each subsection starts with the question in italics and a short answer in a box. Boxes titled *Answer from general knowledge (not in the slides)* contain standard textbook material; numerical values are **typical orders of magnitude**. References to "Unit n" point to the main notes (`Apuntes_Tecnologia_Aeroespacial.pdf`).

1.1 Mass estimate for single-stage and two-stage launch vehicles

Question 13. Carry out a mass estimation exercise for a single-stage-to-orbit and a two-stage-to-orbit launch vehicle.

SHORT ANSWER
With $\Delta v \approx 9.4$ km/s and $I_{sp} = 350$ s, a single stage needs a mass ratio $m_0/m_f = 15.5$ and its structure must be lighter than 6.5 % of the propellant plus structure: it is marginal even with a very optimistic structure ($\varepsilon = 0.05$ leaves a payload of 1.5 % of the launch mass). Two stages need only a ratio of 3.9 each and, with a realistic $\varepsilon = 0.10$, give a payload of 2.9 %. Staging makes orbit feasible.

Model and assumptions (Tsiolkovsky equation of Unit 6 of the main notes; own calculation):

- Required $\Delta v \approx 9.4$ km/s: 7.8 km/s of orbital speed plus about 1.5 to 1.6 km/s of losses (gravity, drag, steering, partly offset by the Earth's rotation). This is a typical order of magnitude, the value of Unit 6 of the main notes.
- Same engine in all the stages with $I_{sp} = 350$ s ($v_e = 3\,434$ m/s); in reality the first stage has lower I_{sp} at sea level and the upper stage a higher one in vacuum.
- **Structural coefficient** $\varepsilon = m_s/(m_s + m_p)$: structure (tanks, engines) over structure plus propellant of the stage. Typical order of magnitude for liquid stages: 0.06 to 0.12. The same ε for all the stages, and the same Δv for each (the optimum split for identical stages).

Equations. For a stage with payload m_{pl} (on top of it): $m_0 = m_s + m_p + m_{pl}$, $m_f = m_s + m_{pl}$, $m_s = \varepsilon(m_0 - m_{pl})$. With the mass ratio $R = m_0/m_f = e^{\Delta v/(I_{sp} g_0)}$ the payload fraction is

$$\lambda = \frac{m_{pl}}{m_0} = \frac{1/R - \varepsilon}{1 - \varepsilon}, \qquad \lambda_{total} = \lambda_1 \, \lambda_2 \cdots$$

and the stage is possible only if $1/R > \varepsilon$.

SINGLE STAGE VERSUS TWO STAGES, PAYLOAD 10 T (OWN CALCULATION)
Single stage. $R = e^{9400/3434} = 15.45$, $1/R = 0.0647$. With $\varepsilon = 0.10$ (typical): $1/R < \varepsilon$, so it is **impossible** (even with no payload the structure alone is heavier than the allowed final mass). With the very optimistic $\varepsilon = 0.05$: $\lambda = (0.0647 - 0.05)/0.95 = 1.5\,\%$, so $m_0 = 10/0.0155 = 645$ t, 94 % of it propellant. With $\varepsilon = 0.06$: $\lambda = 0.5\,\%$, $m_0 = 2\,000$ t.
Two stages, $\Delta v = 4.7$ km/s each. $R = e^{4700/3434} = 3.93$, $1/R = 0.254$. With $\varepsilon = 0.10$:

$$\lambda_1 = \lambda_2 = \frac{0.254 - 0.10}{0.90} = 0.1716, \qquad \lambda_{total} = 0.1716^2 = 2.9\,\%$$

Masses from the top: $m_{0,2} = 10/0.1716 = 58.3$ t (propellant $0.9\,(58.3 - 10) = 43.5$ t, structure 4.8 t); $m_{0,1} = 58.3/0.1716 = 340$ t (propellant $0.9\,(340 - 58.3) = 253$ t, structure 28 t). Total

launch mass **340 t** for 10 t of payload, against 645 t for the (almost impossible) single stage. For comparison, hydrogen/oxygen with $I_{sp} = 450$ s and $\varepsilon = 0.08$ gives a single-stage payload of $(0.1189 - 0.08)/0.92 = 4.2\,\%$: this is why single-stage-to-orbit designs aim at hydrogen, and still have been very hard to build.

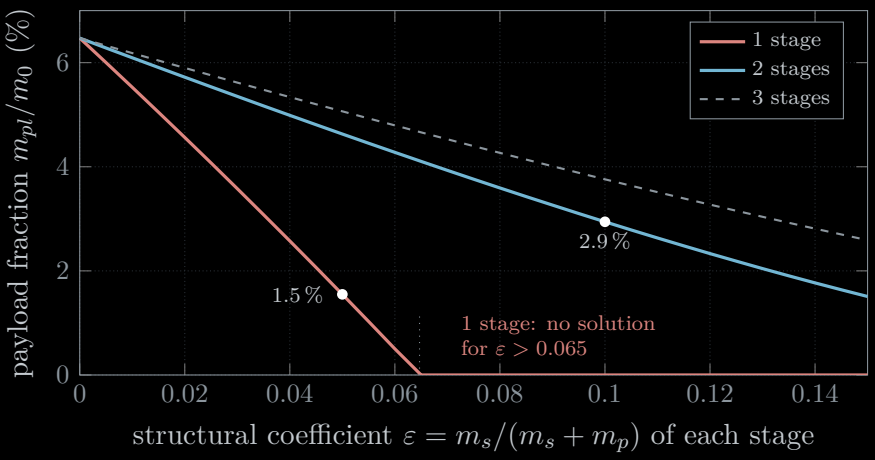


Figure 8.1. Payload fraction m_{pl}/m_0 against the structural coefficient ε of each stage, for $\Delta v = 9.4$ km/s and $I_{sp} = 350$ s (same ε and Δv per stage). A single stage disappears at $\varepsilon = 0.065$; with several stages the same technology can reach orbit. The dots are the two examples of the text. Own calculation.

Reading the result. At $\varepsilon = 0$ all the options are equal (a 6.5 % payload fraction, the $1/R_{total}$ of the whole mission), because staging only helps by discarding dead mass. The higher ε is, the more staging pays; beyond three stages the gain is small compared with the added complexity (the Tsiolkovsky figure of Unit 6 in the main notes shows the propellant fraction of a single stage).

1.2 Secondary radar, TCAS and ADS-B

Question 16. What are the principles of operation of secondary surveillance radar (SSR), TCAS and ADS-B? What is each used for, and how do they differ?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
All three rely on the aircraft **transponder**, a radio on board. **SSR** (ground radar): the ground station interrogates (1030 MHz) and the transponder replies (1090 MHz) with an identity code, altitude or address: it is used by air traffic control to know who and where each aircraft is. **TCAS**: an aircraft interrogates the transponders of nearby aircraft with the same frequencies, to avoid collisions without ground help (traffic and resolution advisories). **ADS-B**: the aircraft itself broadcasts its GNSS position and velocity periodically, without being interrogated; it is received by ground stations and by other aircraft. The air-traffic-control figure of Unit 1 of the main notes shows who controls the aircraft in each phase; the navigation-aids figure of Unit 1 the ground navigation aids.

Primary versus secondary radar. A *primary* radar sends a pulse and listens for the echo reflected by the aircraft: it needs no cooperation, but it only gives range and bearing and the echo is weak. A *secondary* radar (SSR) adds a cooperative element: the ground antenna, usually mounted on top of the primary one, transmits an **interrogation** pulse pair at **1030 MHz**; the aircraft transponder answers on **1090 MHz** with a coded message. Range comes from the reply delay, bearing from the antenna direction. The **modes** define what is asked: **Mode A** (an identity code of four octal digits, 4 096 codes, the *squawk*), **Mode C** (pressure altitude, in steps of 100 ft), and **Mode S** (selective: each aircraft has a unique 24-bit address, so it can be interrogated individually, and data such as flight identification can be exchanged).

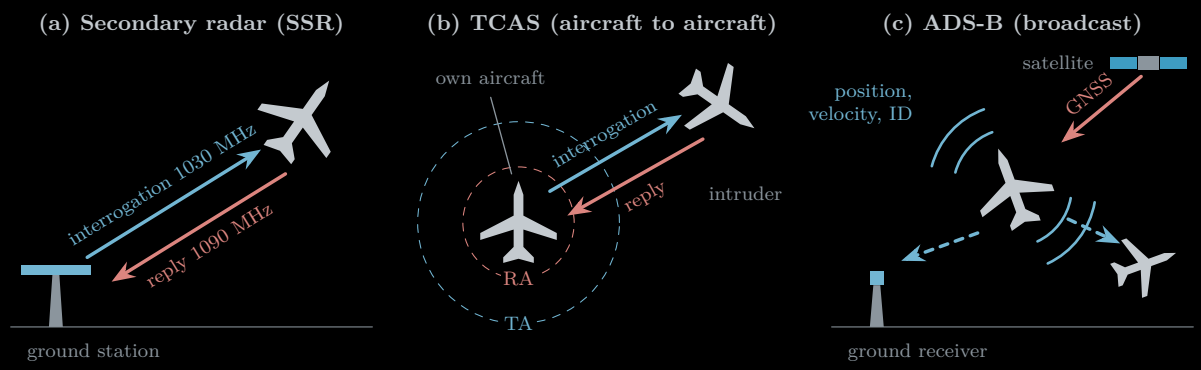


Figure 8.2. Three surveillance systems that use the transponder (schematic). (a) The ground SSR interrogates and the aircraft replies. (b) The own aircraft's TCAS interrogates the intruder's transponder; the dashed circles represent the alert zones (TA outside, RA inside). (c) ADS-B: the aircraft broadcasts what its GNSS receiver computes and nobody needs to ask.

TCAS (traffic alert and collision avoidance system; ACAS in ICAO terms). Its logic is on board. It interrogates on 1030 MHz and listens to the replies on 1090 MHz of every transponder nearby (modes A/C and S), and computes range, closing rate and relative altitude of each *intruder*. If the estimated time to the closest point of approach (the *tau*) falls below a threshold (typically about 40 s for a **traffic advisory, TA**, and 25 s for a **resolution advisory, RA**; thresholds depend on altitude), it warns the pilot (TA) and then orders a vertical manoeuvre (RA: "climb", "descend", "adjust vertical speed"). If the intruder also has TCAS, both aircraft coordinate their advisories through Mode S, so that one climbs and the other descends. It is mandatory on large transport aircraft. It only sees aircraft that have a working transponder.

ADS-B (automatic dependent surveillance–broadcast). *Automatic*: no interrogation is needed; *dependent*: the data come from the aircraft's own navigation system (GNSS). The aircraft broadcasts its identification, position, altitude, velocity and status, about twice per second, on 1090 MHz (extended squitter of Mode S) or on 978 MHz (UAT, used in some regions for general aviation). Any receiver in range uses it: air traffic control (cheap ground stations, even satellites for oceanic coverage) and other aircraft (ADS-B *In*). Precision is that of GNSS, better than that of a radar for position and velocity, and it needs no rotating antenna.

	SSR	**TCAS**	**ADS-B**
Who initiates	ground station interrogates	own aircraft interrogates others	aircraft transmits by itself
Frequencies	1030 MHz up, 1090 MHz down	1030 / 1090 MHz	1090 MHz (or 978 MHz UAT)
Source of position	radar measurement (range, bearing)	relative measurement (range, bearing, altitude)	GNSS on board
Main use	air traffic control surveillance	airborne collision avoidance (last safety net)	surveillance by ATC and awareness of traffic
Who uses it	controllers	pilots (TA/RA)	controllers and pilots
Needs	ground radar network	transponder in the other aircraft	GNSS and a transmitter on board; receivers
Limitations	cost of radars, limited coverage, only cooperative targets	only transponder-equipped targets; vertical advisories	depends on GNSS; data are not authenticated

In short: SSR and TCAS *ask*, ADS-B *tells*; SSR serves the controller on the ground, TCAS serves the pilot with no controller involved, and ADS-B serves both. None of them detects an aircraft with its transponder off; only the primary radar does.

1.3 The rotor in a few lines (introduction for Questions 18 to 20)

The questions on helicopters use some ideas that are not in the units; this short summary (general knowledge) is enough to follow them.

- The **main rotor** is a set of rotating wings (blades) that produces both lift and propulsion. The thrust is changed with the **collective pitch**, which varies the pitch of all the blades together (the lever in the pilot's left hand).
- With **cyclic pitch** the pitch of each blade varies with its azimuth in each turn, which tilts the rotor disc and so the thrust vector: forward, backward or sideways (the stick in the right hand). A helicopter does not tilt its fuselage to go forward; it tilts the rotor (Figure 8.3b).
- **Torque reaction**: the engine drives the rotor and the fuselage reacts with an opposite torque, which a **tail rotor** (or another device) must cancel; the pedals control it and with it the yaw.
- **Dissymmetry of lift**: in forward flight the blade that advances sees a speed $\Omega R + V$ and the retreating one $\Omega R - V$ (Figure 8.3a), so with equal pitch the lift would be much larger on the advancing side and the helicopter would roll. It is cured by **blade flapping** (the blades are hinged or flexible and rise on the advancing side and descend on the retreating side, which reduces and increases their angle of attack) and by the cyclic pitch. Near the hub, the retreating blade sees reverse flow.

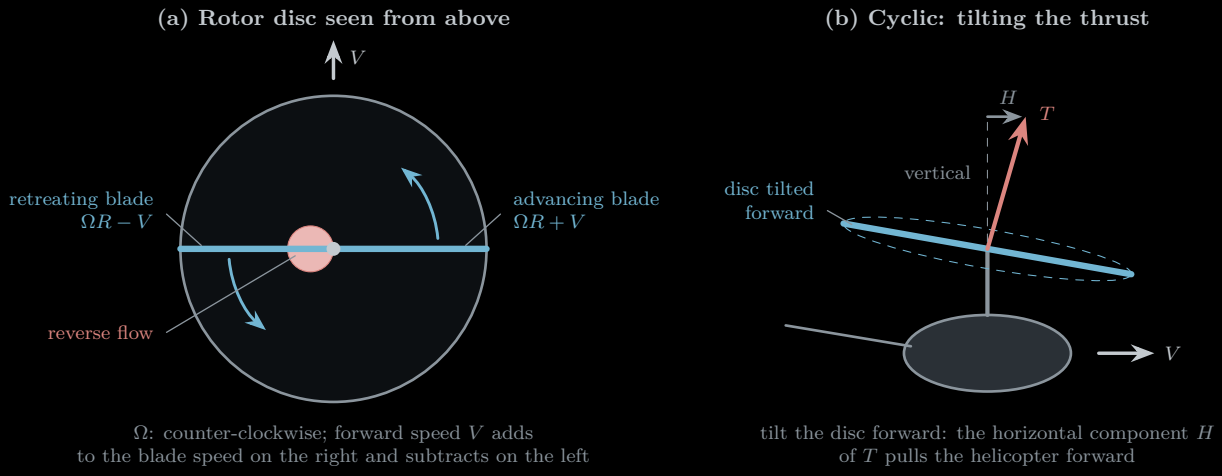


Figure 8.3. (a) Rotor disc seen from above in forward flight at speed V: the advancing blade (right) moves faster relative to the air than the retreating blade (left); a small circle near the hub is the reverse-flow region. (b) Side view: the cyclic tilts the disc and the thrust T forward, giving a horizontal component H. Schematic.

SPEEDS ON THE DISC (ILLUSTRATIVE NUMBERS)
Rotor radius $R = 5.5$ m at 400 rpm: $\Omega = 41.9$ rad/s and $\Omega R \approx 230$ m/s. In forward flight at $V = 60$ m/s (117 kt): advancing tip $230 + 60 = 290$ m/s (Mach 0.85 at sea level, $a = 340$ m/s), retreating tip $230 - 60 = 170$ m/s. The reverse-flow region is a circle of diameter $V/\Omega = 1.4$ m on the retreating side. This is why the top speed of conventional helicopters is limited (Question 19).

1.4 Vertical flight of a helicopter: climb and descent

Question 18. Have a look at the document on vertical flight of helicopters: what happens when they climb or descend at different speeds?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
The "vertical flight" document is not available in the files used for these notes, so the answer comes from the standard **momentum theory** of the rotor. In hover the rotor induces a velocity $v_h = \sqrt{T/(2\rho A)}$ through the disc. **In climb** the induced velocity *decreases* (and power is needed to climb); **in slow descent** (about 0 to $2v_h$) the rotor sinks into its own wake and enters the **vortex ring state**, with loss of thrust and control; **in fast descent** (above about $2v_h$) the air passes upward through the disc (**windmill brake state**) and the rotor can turn by itself: **autorotation**.

Momentum theory. Model the rotor as an actuator disc of area $A = \pi R^2$ that accelerates the air; the induced velocity at the disc is v_i, and far downstream the wake velocity is $2v_i$ (in hover). The vertical speed of the helicopter is V_c (positive upward). The mass flow through the disc is $\dot{m} = \rho A\,(V_c + v_i)$ and the thrust is the momentum change, $T = \dot{m} \cdot w$ with w the velocity increment $2v_i$:

$$T = 2\rho A\,(V_c + v_i)\,v_i \;\Rightarrow\; v_h = \sqrt{\frac{T}{2\rho A}}\;(V_c = 0), \qquad \frac{v_i}{v_h} = -\frac{V_c}{2v_h} + \sqrt{\left(\frac{V_c}{2v_h}\right)^2 + 1}$$

The ideal power is $P = T\,(V_c + v_i)$: the sum of the climb power TV_c and the induced power Tv_i. For descent ($V_c < 0$) the same formula holds only in the windmill brake state, written for $V_c \le -2v_h$ as $v_i/v_h = \dfrac{|V_c|}{2v_h} - \sqrt{\left(\dfrac{V_c}{2v_h}\right)^2 - 1}$.

Flight state	V_c/v_h	**What happens**
Climb	> 0	the air arrives from below with speed V_c, so for the same thrust v_i is smaller than in hover; the power needed is hover power plus climb power
Hover	0	$v_i = v_h$; the most basic reference ($P_{ideal} = Tv_h$)
Vortex ring state (and turbulent wake)	about -2 to 0 (slow descent)	the rotor descends at a speed comparable to its own downwash: the flow recirculates around the blade tips in a doughnut-shaped vortex, the thrust becomes unsteady, the helicopter vibrates and sinks faster; **more collective makes it worse**. Momentum theory is not valid; the curve is empirical
Windmill brake state	≤ -2	the flow goes upward through the disc; the rotor extracts power from the air; this is the regime of **autorotation**, used after an engine failure

A LIGHT HELICOPTER (ILLUSTRATIVE NUMBERS)
Mass 1 500 kg ($T = 14\,715$ N), rotor radius 5.5 m ($A = 95.0$ m^2), sea level.

- Hover: $v_h = \sqrt{14\,715/(2 \cdot 1.225 \cdot 95.0)} = 7.95$ m/s; ideal power $Tv_h = 117$ kW (the real one is higher: blade profile drag and other losses).
- Climb at $V_c = 5$ m/s: $v_i = -2.5 + \sqrt{2.5^2 + 7.95^2} = 5.83$ m/s, power $T(V_c + v_i) = 14\,715 \cdot 10.83 = 159$ kW.
- Descent at 4 m/s ($V_c/v_h = -0.5$) or 10 m/s (-1.3): inside the vortex ring band (0 to $-2v_h = 15.9$ m/s). Descent at 16 m/s or more: windmill brake state; at 16 m/s the formula gives $v_i = 7.1$ m/s.

Practical consequences. The vortex ring state arises with a high power setting, a descent rate comparable to v_h and a low forward speed (rule of thumb, order of magnitude: forward speed under about 20 kt); with forward speed the rotor leaves its own wake and the problem disappears.

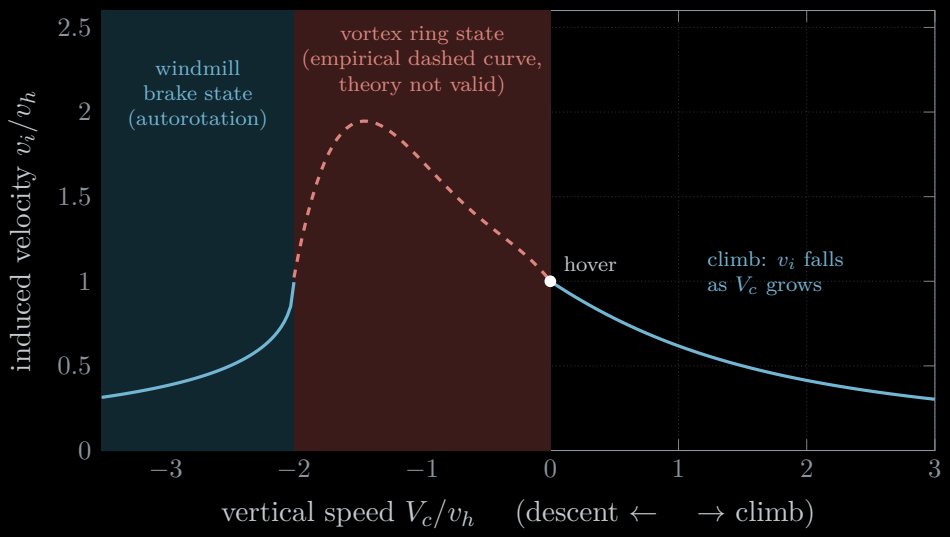


Figure 8.4. Induced velocity over hover value against the vertical speed over the hover velocity (momentum theory in solid lines; the dashed curve is the typical empirical shape in the vortex ring region, drawn from a standard polynomial fit and scaled to equal 1 at hover; only the qualitative shape matters). Climb is on the right, descent on the left. The coloured bands mark the regions.

The recovery is to gain forward speed with the cyclic and lower the collective (never raise it); in autorotation the pilot lowers the collective at once to keep the rotor speed and flares near the ground to use the stored rotor energy.

1.5 Flight envelope of a helicopter

Question 19 (helicopter part). Draw the speed–altitude flight envelope diagram for a helicopter. What are the limits and what are they due to?

> **ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)**
> **Helicopter:** no low-speed limit (it hovers), but a **power ceiling** on top (hover ceiling at zero speed) and a **never-exceed speed** V_{NE} on the right due to **retreating-blade stall** and **advancing-tip compressibility**; at low height and speed lies the **height–velocity "dead man's curve"** where an engine failure cannot be survived.

Helicopter (Figure 8.5).

- **No low-speed limit.** A helicopter can hover, so the left boundary is $V = 0$. The envelope is plotted against altitude (density altitude) to include the effect of temperature.
- **Power ceiling.** Power required exceeds power available above a certain altitude: at $V = 0$ it is the **hover ceiling** (lower out of ground effect than in ground effect). The ceiling is higher at moderate speed, because the induced power falls with forward speed (translational lift) up to the speed of minimum power (about 60 to 80 kt), and then falls again as the parasite power grows with V^3.
- V_{NE} **(never-exceed speed).** The advancing blade tip approaches the speed of sound ($\Omega R + V$, Mach 0.85 to 0.9) and the drag rises (compressibility); the retreating blade has little relative speed ($\Omega R - V$) and needs a large angle of attack to produce its lift, so it **stalls** (vibration, pitch-up, roll). Both worsen with altitude and weight, so V_{NE} is reduced there. Typical maximum speeds of conventional helicopters are 150 to 170 kt, an order of magnitude.
- **Height–velocity ("dead man's") diagram.** Panel (b) shows in red the combinations of height above ground and speed to avoid: if the engine fails at low speed and high height (above the low hover) there is not enough time or speed to build up rotor speed and flare, and

if it fails at high speed and very low height there is no room to flare. The exact curve is in the flight manual of each helicopter (typical orders of magnitude: low hover under about 15 ft; avoid area up to 400 to 500 ft near zero speed).

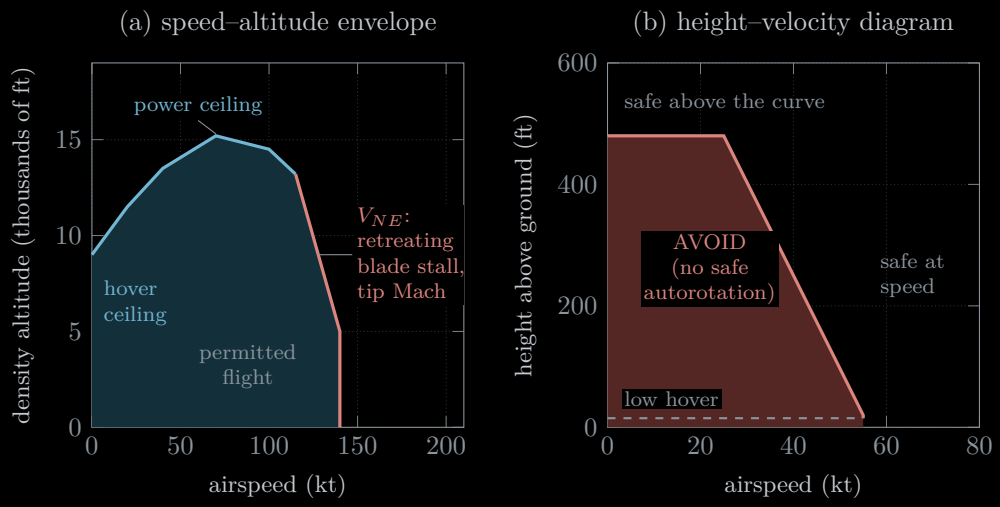


Figure 8.5. (a) Speed–altitude envelope of a conventional helicopter (illustrative). (b) Height–velocity diagram of a single-engine helicopter (illustrative shape): the red region is to be avoided. Both are schematic and the numbers are only typical orders of magnitude.

1.6 Yaw control of a tandem twin-rotor helicopter

Question 20. How do you control yaw in a tandem twin-rotor helicopter?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
By **differential lateral cyclic pitch**: the two rotors are tilted sideways in opposite directions, so the front rotor pushes to one side and the rear rotor to the other. The two lateral forces, separated by the fuselage length, give a **yawing moment**. There is no tail rotor: the pedals command this differential cyclic. In a tandem the rotors turn in opposite directions, so their torques cancel; **pitch** is controlled by **differential collective** (more thrust on one rotor than on the other), and **roll** by lateral cyclic in the same direction on both rotors.

Explanation. Each rotor has its own swashplate. For the pedals, the lateral cyclic of the front rotor is tilted, for example, to the right and that of the rear rotor to the left by the same amount. The net lateral force is zero (the aircraft does not move sideways) but the two forces F_f and F_a act at a distance L from the centre of gravity, so the yawing moment is $M_z \approx (F_f + F_a)\, L/2$, in magnitude. For the other axes: the **collective lever** raises or lowers the collective on both rotors equally (climb); the **cyclic stick** forward–back tilts both discs the same way (forward flight) and sideways tilts them the same way (roll); the **pedals** give the differential lateral cyclic.

Compared with a single rotor. The tail rotor of the conventional layout takes power that does not contribute to lift, and the tandem uses all the power for lift (**no anti-torque losses**); it also allows a wide centre-of-gravity range and a large cabin. The disadvantages are the complexity of two rotors with their transmission, and the rear rotor working in the wake of the front one (loss of efficiency and vibration). Other configurations that avoid a tail rotor, such as coaxial rotors, also cancel torque with counter-rotating rotors.

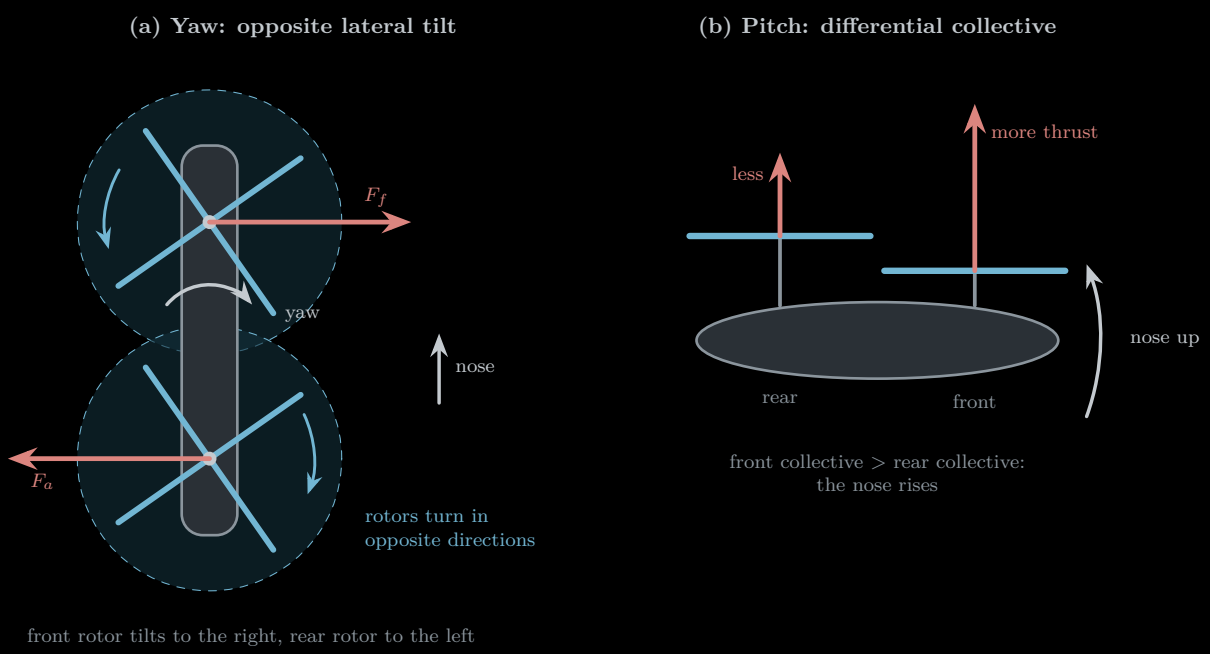


Figure 8.6. Tandem rotor (Chinook type). (a) Top view: the front rotor tilts its thrust to the right and the rear rotor to the left; the couple turns the nose to the right. (b) Side view: more collective on the front rotor than on the rear raises the nose. Schematic.